

Lab 04 haskell

Ejercicio 1.

```
[13]: miLength :: [a] -> Int
      miLength [] = 0
      miLength (_:xs) = 1 + miLength xs
      miLength [2,4,6,8,10]
```

5

```
[12]: miLast :: [a] -> a
      miLast [x] = x
      miLast (_:xs) = miLast xs
      miLast [1,2,3,4,5,6]
```

6

```
[11]: duplicar :: [Int] -> [Int]
      duplicar xs = map (* 2) xs
      duplicar [1,12,3,4,5,6]
```

Eta reduce

Found:

Why Not:

```
duplicar xs = map (* 2) xs
duplicar = map (* 2)

[2,24,6,8,10,12]
```

```
eliminarPares :: [Int] -> [Int]
eliminarPares xs = filter odd xs
eliminarPares [1,2,3,4,5,6,7,8,9,10]
```

Eta reduce

Found:

Why Not:

```
eliminarPares xs = filter odd xs
eliminarPares = filter odd

[13,15,17,19]
```

Ejercicio 2

1.

```
[x^2 | x <- [1..10]]
```

```
[1,4,9,16,25,36,49,64,81,100]
```

```
[x | x <- [1..30], mod x 3 == 0]
```

```
[3,6,9,12,15,18,21,24,27,30]
```

```
[ (x, y) | x <- [1..5], y <- [1..5], x<y]
```

```
[(1,2),(1,3),(1,4),(1,5),(2,3),(2,4),(2,5),(3,4),(3,5),(4,5)]
```

2.

```
divisores :: Int -> [Int]
divisores n = [x | x <- [1..n], n `mod` x == 0]
divisores 10
```

```
[1,2,5,10]
```

3.

```
esPerfecto :: Int -> Bool
esPerfecto n = sum [x | x <- [1..n-1], n `mod` x == 0] == n
esPerfecto 6
```

```
True
```

Ejercicio 3

Ejercicio 4

```
]:: procesarLecturas :: (Double -> Double) -> [Double] -> [Double]
procesarLecturas f xs = map f xs

celsiusAFaranheit :: Double -> Double
celsiusAFaranheit c = c * 9 / 5 + 32

crearFiltro :: (a -> Bool) -> ([a] -> [a])
crearFiltro p = filter p

soloPares :: [Int] -> [Int]
soloPares = crearFiltro even

enRango :: Int -> Int -> (Int -> Bool)
enRango minimo maximo = \x -> x >= minimo && x <= maximo

entre5y10 :: Int -> Bool
entre5y10 = enRango 5 10

potenciador :: Int -> (Int -> Int)
potenciador n = \x -> x ^ n

potencia5 :: Int -> Int
potencia5 = potenciador 5
```

Esta reduce

```
[22]: procesarLecturas celsiusAFaranheit [0,10,25]  
[32.0,50.0,77.0]
```

```
[23]: soloPares[1..20]  
[2,4,6,8,10,12,14,16,18,20]
```

```
[25]: entre5y10 7  
True
```

```
[26]: potencia5 7  
16807
```

```
[ ]:
```